

Practical 9

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In [19]: # — Rail Fence Cipher —————  
  
'''  
Logic of Rail Fence Cipher  
  
Rail 0 : H . . . O . . . D  
Rail 1 : . E . L . W . L .  
Rail 2 : . . L . . . R . .  
  
Read across → H O D + E L W L + L R = HODELWLLR  
  
'''  
  
# Step 1: Encrypt  
def encrypt(text, rails):  
    fence = [[] for _ in range(rails)]  
    rail, direction = 0, 1  
  
    for char in text:  
        fence[rail].append(char)  
        if rail == 0:            direction = 1  
        if rail == rails - 1:    direction = -1  
        rail += direction  
  
    return ''.join(''.join(row) for row in fence)
```

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In [17]: # Step 2: Decrypt  
def decrypt(text, rails):  
    n = len(text)  
    pattern = []  
    rail, direction = 0, 1  
  
    for i in range(n):                # track which rail each position belongs to  
        pattern.append(rail)  
        if rail == 0:                direction = 1  
        if rail == rails - 1:        direction = -1  
        rail += direction  
  
    indices = sorted(range(n), key=lambda i: pattern[i])  
    result = [''] * n  
    for i, char in zip(indices, text):  
        result[i] = char  
  
    return ''.join(result)
```

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In [18]: # — Demo —————  
msg      = "HELLOWORLD"  
rails    = 3
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enc = encrypt(msg, rails)
dec = decrypt(enc, rails)

print("=== Rail Fence Cipher ===")
print(f"Original : {msg}")
print(f"Rails      : {rails}")
print(f"Encrypted  : {enc}")
print(f"Decrypted  : {dec}")
```

```
=== Rail Fence Cipher ===
Original : HELLOWORLD
Rails    : 3
Encrypted : HOLELWRDLO
Decrypted : HELLOWORLD
```